

Cross-Framework Implementation of Fetal Brain MRI Plane Classification Model: Comparative Inference Memory Consumption Between Deep Learning Frameworks

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Abstract—Fetal brain development is a highly intricate process that is susceptible to various congenital anomalies, making Magnetic Resonance Imaging (MRI) a vital tool for prenatal assessment. Deep learning models have emerged as powerful techniques for detecting fetal brain abnormalities from MRI images; however, their computational demands necessitate efficient memory utilization to ensure scalability and real-time applicability. This study presents a cross-framework comparison of memory utilization, inference speed, and prediction accuracy for fetal brain MRI classification using different deep learning frameworks: TensorFlow, PyTorch, and ONNX. The evaluation employs both the VGG16 convolutional neural network (CNN) and a Vision Transformer (ViT) model to assess the impact of framework selection on performance and resource efficiency. Results indicate that TensorFlow provides a balanced trade-off between model size and inference speed, ONNX enhances cross-framework compatibility with efficient storage but exhibits slower inference, while PyTorch achieves the highest accuracy and fastest inference at the expense of greater memory consumption. The inclusion of ViT further highlights the adaptability of transformer-based models in medical imaging tasks, demonstrating competitive performance across frameworks. These findings offer valuable insights for optimizing deep learning models in prenatal diagnostics, ensuring both high diagnostic accuracy and efficient computational resource management.

Keywords— *Fetal Brain MRI, Cross-Framework Comparison, Resource-Efficient AI, Deep Learning, Memory Utilization*

I. INTRODUCTION

Fetal brain development is a complex and intricate process, vulnerable to a range of congenital anomalies. Magnetic Resonance Imaging (MRI) plays a crucial role in assessing fetal brain anatomy and detecting abnormalities prenatally [1]. MRI imaging provides non-invasive, high-resolution insights into fetal brain development across multiple views, including axial, coronal, and sagittal planes [2]. The use of deep learning models for detecting fetal brain anomalies from MRI images has recently emerged as a popular and compelling research area among practitioners [3], [4]. However, the computational demands of deep learning models for medical image analysis necessitate efficient memory utilization to ensure scalability, reliability, and real-time applicability [5]. Modern

deep learning frameworks such as TensorFlow [6], PyTorch [7], and Open Neural Network Exchange (ONNX) [8] offer powerful capabilities for model development and deployment. Each framework, however, has distinct characteristics in terms of memory consumption, inference speed, and optimization strategies, making it essential to understand their performance trade-offs when applied to high-resolution medical imaging tasks.

Efficient memory utilization is critical when deploying deep learning models in resource-constrained environments such as edge devices, mobile applications, or cloud-based diagnostic platforms [9]. Large model sizes and high inference times can hinder real-time decision-making, limiting the feasibility of AI-driven fetal brain MRI analysis in clinical settings [10]. Optimizing memory consumption without sacrificing model accuracy is key to ensuring smooth integration into medical workflows. Thus, an in-depth comparative study of the memory footprint and computational efficiency across different deep learning frameworks is essential for guiding model selection and deployment strategies. Despite the growing adoption of AI in medical imaging, there is limited research comparing the memory efficiency of deep learning frameworks when applied to fetal brain MRI analysis. Existing studies primarily focus on accuracy and inference speed, often overlooking the impact of model size and memory footprint on real-world deployment. Given the substantial variations in memory utilization across TensorFlow, PyTorch, and ONNX, a comprehensive comparison is necessary to identify the most suitable framework for fetal brain MRI classification tasks.

Therefore, this research aims to conduct a cross-framework comparison of memory utilization for fetal brain MRI classification using deep learning techniques. This work investigates and compares the memory utilization and performance of a fetal brain MRI prediction model implemented across three popular deep learning frameworks: TensorFlow, PyTorch, and ONNX. It specifically focuses on evaluating the impact of framework choice and deployment strategy on memory consumption, inference speed, and prediction accuracy. To

achieve a comprehensive analysis, both convolutional neural networks (CNN), represented by the VGG16 model [11], and transformer-based architectures, such as Vision Transformer (ViT) pretrained model [12], are employed for comparative evaluation. By analyzing these factors, this study seeks to identify the most memory-efficient approach for deploying fetal brain MRI prediction models in various computational environments, as well as ensuring efficient real-life implementation of the predictive models. Thus, the result analysis provides a crucial reference for researchers and practitioners seeking to optimize deep learning models for fetal brain MRI analysis while ensuring efficient memory usage with high diagnostic accuracy.

The remainder of this paper is organized as follows: Section 2 covers Literature Review discussing existing research on this domain; Section 3 includes methodology which covers the methodological overview, the dataset characteristics, training procedures, and experimental setup for framework comparisons; Section 4 highlights Results and Discussion presenting the comparative analysis of across the frameworks; and finally Section 5 includes Conclusion summarizing the key takeaways, limitations, and suggests potential areas for future research.

II. LITERATURE REVIEW

The application of deep learning to fetal brain MRI has become increasingly prevalent due to its ability to automatically learn complex patterns and features from imaging data [13]. Traditional methods of image analysis often require manual feature engineering, which can be time-consuming and subjective. Deep learning, particularly Convolutional Neural Networks (CNNs), offers a powerful alternative by automatically extracting relevant features from the raw medical image data [14]. This has led to significant advancements in various tasks related to fetal brain MRI analysis.

Such as, Shen et al. [15] developed an attention-guided deep learning model for predicting gestational age using fetal brain MRI, achieving high accuracy with an R^2 score of 0.945 and a mean absolute error of 6.7 days. Fidon et al. [16] introduced a distributionally robust optimization approach, improving fetal brain MRI segmentation accuracy, particularly for pathological cases like spina bifida. Zhao et al. [17] presented a 3D U-Net model for automated fetal brain segmentation, outperforming traditional methods with an average Dice score of 0.897. Sreelakshmy et al. [18] proposed ReU-Net for cerebellum segmentation from fetal ultrasound images, achieving a Dice score of 91%.

However, while many studies have successfully applied deep learning models to fetal brain imaging tasks, few have analyzed critical factors such as memory utilization, inference time, and computational efficiency. In deep learning, these aspects are crucial, as models often require significant computational resources and time to process complex medical images. Despite their impressive accuracy, the efficiency of these models—particularly in terms of inference speed and memory consumption—has been largely overlooked.

Nevertheless, few researchers have focused on memory utilization and optimization in case of deep learning models in other research domains. Such as, Meng et al. [19] focused on optimizing GPU memory utilization for deep learning by introducing a swap-out/in strategy that offloads feature maps from GPU memory to host memory, effectively increasing available memory for training deep models. Their method integrates seamlessly into TensorFlow, allowing larger batch sizes and deeper models to be trained without modifying model descriptions. In another work, Kim et al. (2020) [20] proposes a multi-GPU memory optimization strategy to enhance deep learning training performance. Their approach includes intelligent prefetching from CPU to GPU and an optimized swap-in/out mechanism, addressing I/O bottlenecks in shared interconnects like PCIe and NVLink. They integrate their method into TensorFlow and show significant performance improvements, particularly in multi-GPU environments. Also, Steiner et al. [21] introduces MODEL, an ILP-based algorithm that optimizes tensor lifetimes and memory locations to reduce peak memory usage in deep learning models. Unlike existing methods that trade memory for computation, MODEL minimizes memory consumption without sacrificing accuracy or increasing training time. Their method achieves an average memory reduction of 30% across various deep learning architectures. Moreover, Carneiro et al. [22] conducted a performance analysis of Google Colaboratory, emphasizing its utility as a cloud-based Jupyter Notebook environment for accelerating deep learning applications. They highlighted Colab's provision of pre-configured runtimes and free access to high-performance GPUs, making it a valuable tool for machine learning research and education.

However, though researchers have investigated memory optimization analysis in deep learning models in various domains, there has been limited attention given to memory utilization and optimization in healthcare analytics, particularly when using medical image data. Additionally, very few studies have analyzed performance in the context of transformer-based models, despite their growing success in medical image analysis. Therefore, this study focuses on addressing these research gaps in terms of fetal brain MRI image classification using deep learning techniques with both CNN and transformer models on multiple frameworks.

III. METHODOLOGY

The research methodology that has been applied in this study can be divided into multiple steps, such as data collection, data analysis and preprocessing, applying deep learning models in multiple frameworks, and finally conducting comparative performance analysis to investigate the memory utilization by the frameworks. The methodological overview has been illustrated in Figure 1. A detailed explanation of each step is provided in the following subsections.

A. Data Collection

The dataset used in this research work consists of 741 developmentally normal fetal brain MRI scans from the Lucile

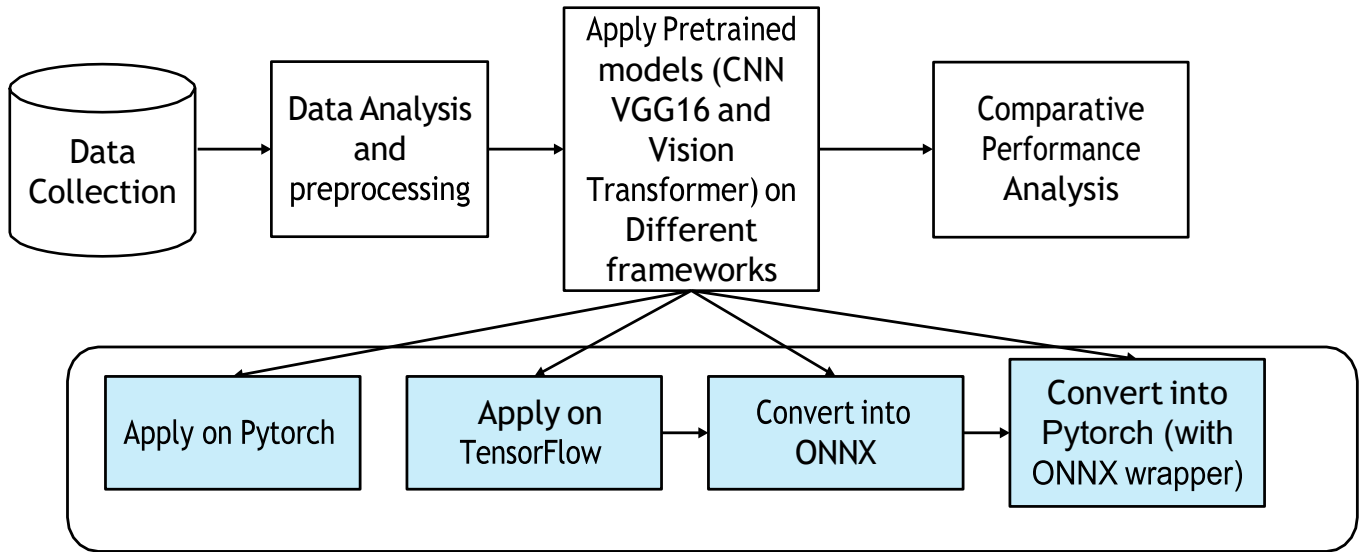


Fig. 1: Methodological Overview

Packard Children’s Hospital, with corresponding gestational ages ranging from 19 to 39 weeks [23]. The images, available in axial, sagittal, and coronal planes, are stored in JPEG format and organized into folders. Each MRI sequence is accompanied by a label file containing patient IDs, gestational ages, and the specific plane sequences. This diverse dataset is valuable for building deep learning models aimed at predicting gestational age and supporting neurodevelopmental assessment.

B. Data Analysis and Preprocessing

Data preprocessing is a crucial step in ensuring high-quality input for model training and evaluation. The dataset consists of MRI images of fetal brains, which are loaded, processed, and prepared for training through various transformation techniques. This preprocessing pipeline plays a fundamental role in standardizing the data and ensuring consistency across different frameworks such as TensorFlow, PyTorch, and ONNX.

The images are loaded using OpenCV (cv2) as grayscale images and subsequently converted to RGB using `cv2.cvtColor(img, cv2.COLOR_BGR2RGB)` to align with deep learning models expecting three-channel inputs. After conversion, images are resized to 224×224 pixels to ensure uniformity. The dataset is then split into training and testing sets using an 80-20 split. To normalize the pixel values, the image arrays are scaled by dividing by 255, bringing pixel values into the [0,1] range, which enhances model convergence.

Through these preprocessing steps, the dataset is efficiently prepared for comparative analysis across TensorFlow, PyTorch, and ONNX. This approach ensures consistency across frameworks, allowing for a fair assessment of memory utilization and performance while training deep learning models for fetal brain MRI prediction.

C. Applying Pretrained Models on Different Frameworks

This research applies both the VGG16 model and the Vision Transformer (ViT) model from Hugging Face to evaluate their effectiveness in fetal brain MRI classification.

- **VGG16 Model:** The VGG16 model has been implemented across multiple frameworks for comparative analysis. In TensorFlow, the pre-trained VGG16 model (with ImageNet weights) has been used, excluding the fully connected layers (`include_top=False`). A Global Average Pooling (GAP) layer and fully connected layers have been added, leading to a final softmax classification layer. The model has been trained for 30 epochs with the Adam optimizer, using sparse categorical cross-entropy loss.
- **ONNX Conversion and PyTorch Integration:** The trained TensorFlow model was converted into the ONNX format using `tf2onnx` to enable interoperability. The ONNX model was subsequently integrated into PyTorch using a custom wrapper for inference.
- **Native PyTorch Implementation:** A fully native PyTorch implementation of VGG16 was trained using the `torchvision.models` package. Training included forward propagation, loss computation (cross-entropy loss), and optimization using Adam.
- **Vision Transformer (ViT) Model from Hugging Face:** In addition to VGG16, a Vision Transformer (ViT) model has been applied to investigate the effectiveness of transformer-based architectures in fetal MRI classification. The `ViTForImageClassification` model from the Hugging Face `transformers` library was fine-tuned using the fetal brain MRI dataset. The pre-trained ViT model, initialized with ImageNet weights, was modified by adding custom classification layers for

fetal MRI categorization. Fine-tuning was performed using the AdamW optimizer with a learning rate of $5e-5$, training for 20 epochs. To ensure a comprehensive memory utilization analysis, the ViT model was implemented across PyTorch, ONNX, and TensorFlow frameworks, similar to the VGG16 model, allowing for a comparative assessment of resource efficiency across different deep learning environments.

D. Computational Resources

The entire implementation was performed on Google Colab using the T4 GPU. Google Colab's CUDA-enabled T4 GPU significantly accelerated model training and inference by optimizing tensor computations. The models were executed with mixed-precision training where applicable to improve performance and reduce memory consumption.

E. Comparative Performance Analysis

For performance evaluation, each framework and model was assessed using three key metrics: classification reports, inference time, and model file size. The classification report from scikit-learn provides an evaluation of precision, recall, and F1-score. Additionally, a confusion matrix was generated to visualize true positives, false positives, and false negatives.

To evaluate inference speed, the time taken for inference was measured by recording timestamps before and after model execution. For PyTorch, class predictions were extracted using `torch.argmax()`. Finally, trained model file sizes were measured to compare storage efficiency across frameworks.

These evaluations provide a holistic view of model effectiveness, efficiency, and resource demands across different deep learning frameworks.

IV. RESULT ANALYSIS

The comparative analysis of the VGG16 and ViT models across TensorFlow, PyTorch, and ONNX frameworks reveals significant insights into their performance, memory utilization, and inference efficiency. The results, as presented in Table 1, highlight the trade-offs between model size, inference time, and accuracy, which are critical for deploying deep learning models in resource-constrained environments such as medical imaging applications. Additionally, Figure 2 shows the training accuracy and loss of both models over 30 epochs.

A. VGG16 Model Performance

First, the TensorFlow implementation of the VGG16 model demonstrates a balance between model size and inference speed. With a model size of 62.27MB, TensorFlow achieves an average inference time of 77.77ms per test image while maintaining a high test accuracy of 0.946. This performance is particularly noteworthy given TensorFlow's widespread use in production environments, where its optimized computational graph and efficient memory management contribute to its reliability and scalability. The relatively small model size and fast inference time make TensorFlow a strong candidate for real-time applications, such as fetal brain MRI analysis, where quick decision-making is essential.

In contrast, the ONNX implementation, while achieving the same test accuracy of 0.946, exhibits a significantly larger inference time of 289.18ms. This increase in inference time can be attributed to the additional overhead introduced by the ONNX runtime, which is designed to provide cross-framework interoperability rather than optimized performance for a specific framework. However, the ONNX model size is slightly smaller at 58.16MB, indicating that the conversion process from TensorFlow to ONNX may have introduced some optimizations in terms of storage efficiency. Despite the slower inference time, ONNX's ability to facilitate cross-framework compatibility makes it a valuable tool for scenarios where model portability across different platforms is a priority.

The PyTorch implementation with the ONNX wrapper presents an interesting case. While it achieves the same accuracy of 0.946, the inference time increases further to 334.49ms, which is the slowest among the three frameworks. This can be attributed to the additional layer of abstraction introduced by the ONNX wrapper, which adds computational overhead during inference. However, the model size is drastically reduced to just 1.18KB, highlighting the efficiency of the ONNX wrapper in terms of storage. This makes the PyTorch-ONNX wrapper approach particularly suitable for environments where storage constraints are critical, but the trade-off in inference speed must be carefully considered.

Finally, the native PyTorch implementation of the VGG16 model achieves the highest test accuracy of 0.948, demonstrating the framework's capability to achieve superior performance when fully leveraging its computational graph. However, this comes at the cost of a significantly larger model size of 512.22MB, which is substantially larger than the other implementations. The inference time, at 9.17ms per image, is the fastest among all frameworks, making PyTorch an excellent choice for applications where both accuracy and speed are paramount, but memory and storage constraints are less critical.

B. ViT Model Performance

The Vision Transformer (ViT) model, known for its superior feature extraction capabilities in image classification, was also evaluated across TensorFlow, PyTorch, and ONNX frameworks. The results indicate that ViT models tend to have larger model sizes compared to CNN-based architectures like VGG16, but they offer faster inference times in certain implementations.

The TensorFlow implementation of ViT has a model size of 85.32MB and an inference time of 95.47ms while achieving a test accuracy of 0.9589. While larger than VGG16, this size is still manageable for real-time applications. The ONNX implementation, however, achieves the same accuracy but with a slightly reduced model size of 79.28MB, though at the cost of an increased inference time of 312.56ms. This slower inference suggests that ONNX may not be as well optimized for transformer-based models as it is for CNNs.

Similar to the VGG16 results, the PyTorch implementation with the ONNX wrapper shows a significant drop in

TABLE I: Comparative Performance Analysis of Memory Utilization between Different Frameworks for transfer-learning with VGG16 and ViT models

Model	Framework	Model Size	Inference Time	Test Accuracy
4*VGG16 (CNN-based Pretrained model)	Tensorflow	62.27MB	77.77ms	0.946
	Onnx	58.16MB	289.18ms	0.946
	Pytorch(Onnx Wrapper)	1.18KB	334.49ms	0.946
	Pytorch	512.22MB	9.17 ms	0.948
4*ViT (Transformer-based pretrained model)	Tensorflow	85.32MB	95.47ms	0.9589
	Onnx	79.28MB	312.56ms	0.9589
	Pytorch(Onnx Wrapper)	1.82MB	375.23ms	0.9589
	Pytorch	675.41MB	7.25ms	0.9589

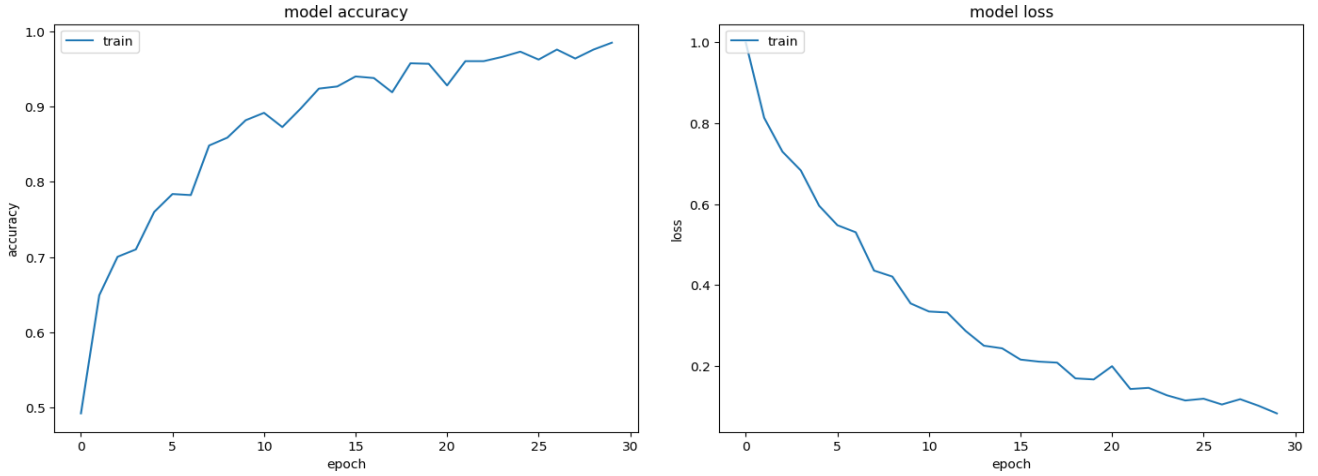


Fig. 2: Model Accuracy and Loss over 30 Epochs (In Tensorflow Framework)

inference efficiency, recording the slowest inference time at 375.23ms, despite having a significantly compressed model size of 1.82MB. This suggests that ONNX's storage optimization benefits may not translate well into efficient inference execution for transformer-based models.

The native PyTorch implementation of ViT stands out with the fastest inference time of 7.25ms, significantly outperforming other implementations. However, this comes at the cost of the largest model size at 675.41MB. This suggests that while PyTorch can optimize transformer models for speed, it requires a much higher memory footprint, making it less suitable for memory-constrained environments.

C. Comparative Analysis

The results indicate that the choice of framework for deploying deep learning models in fetal brain MRI analysis depends heavily on the specific requirements of the application. TensorFlow offers a balanced approach with reasonable model size and inference speed, making it suitable for real-time applications. ONNX provides cross-framework compatibility with efficient storage but at the cost of slower inference. The PyTorch-ONNX wrapper is ideal for storage-constrained environments, though it comes with performance trade-offs. Meanwhile, native PyTorch implementations consistently deliver the fastest inference but require significantly more memory.

Moreover, the results suggest that transformer-based models, like ViT, offer competitive accuracy in fetal MRI classification but require careful framework selection to balance memory usage and inference efficiency. PyTorch provides the fastest inference for ViT, while TensorFlow remains a reliable option with manageable memory usage.

These findings provide valuable guidance for researchers and practitioners aiming to optimize deep learning models for medical imaging tasks, ensuring both high diagnostic accuracy and efficient resource utilization.

V. CONCLUSION

This study provides a comprehensive comparative analysis of memory utilization, inference speed, and accuracy across TensorFlow, PyTorch, and ONNX frameworks for fetal brain MRI classification using both CNN-based (VGG16) and transformer-based (ViT) deep learning models. The results highlight the inherent trade-offs between model size, inference time, and accuracy, offering valuable insights into selecting the most appropriate framework based on specific application requirements. TensorFlow is identified as a balanced choice for real-time applications, offering a good compromise between memory utilization and inference speed. ONNX excels in providing cross-framework compatibility, making it an efficient option for storage while maintaining reasonable performance. PyTorch stands out by achieving the highest accuracy and

the fastest inference speed, though it requires a larger memory footprint, which may be a limiting factor for resource-constrained environments.

However, this study has certain limitations that need to be addressed in future work. Firstly, the evaluation was confined to the VGG16 model and vit model, and different architectures such as ResNet, Inception, EfficientNet, or swin transformer models may exhibit varying performance characteristics. Additionally, the performance metrics were derived from a specific dataset, and the results may vary when applied to other datasets or medical imaging tasks. While the study primarily focuses on memory utilization and inference speed, it does not delve into other optimization techniques, such as quantization or pruning, which could further enhance model efficiency.

Future research aims to extend these findings by evaluating a broader range of deep learning models, including ResNet, Inception, and EfficientNet as well as transformer models to gain a deeper understanding of their memory utilization and performance across the three frameworks. Further investigation into advanced memory optimization techniques, such as quantization, pruning, and knowledge distillation, will be pursued to reduce model size and improve inference speed. Additionally, the impact of hardware acceleration, such as TPUs and FPGAs, on memory utilization and inference speed will be explored to assess how different frameworks perform under varying computational environments. By addressing the current limitations and expanding the scope of this research, future work will provide more comprehensive insights into optimizing deep learning models for medical imaging tasks, ensuring both high diagnostic accuracy and efficient resource utilization.

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